Fundamentals Chapter 1

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1a. Since $1 + (-1) = 0$, $0a = (1 + (-1))a = 1 \cdot a + (-1) \cdot a = a + (-a) = 0$.

1b. Assume for contradiction that $0 = 1$. For all $a \in \mathbb{F}$, $a \cdot 0 = a \cdot 1$. We know that $a \cdot 0 = 0$ and $a \cdot 1 = a$ by the definition of 1, so $a = 0$. But this implies that the only element in $\mathbb{F}$ is 0, which contradicts the definition of a field. Therefore, $0 \neq 1$.

1c. Assume $ab = 0$ and $a, b \neq 0$ so that a^{-1} and b^{-1} exist. Then $aba^{-1}b^{-1} = 0a^{-1}b^{-1}$, so $1 = 0$, which is a contradiction. Therefore at least one of a, b must be 0.

1d. We know that $0a = 0b$. Replacing 0 with $-a + a$ and $-b + b$ gives us

$$a(-b + b) = b(-a + a),$$

$$a(-b) + ab = (-a)b + ab,$$

$$a(-b) + ab + (-ab) = (-a)b + ab + (-ab),$$

$$a(-b) = (-a)b.$$

Also, from $0a = 0$, we can say that

$$0a = 0,$$

$$a(b + (-b)) = 0,$$

$$ab + a(-b),$$

$$ab + (-ab) + a(-b) = -ab,$$

$$a(-b) = -ab.$$

Therefore, $a(-b) = (-a)b = ab$.

2. As shown earlier, $0x = 0$ for all $x \in \mathbb{F}$, and $0 \neq 1$. Therefore, $x \cdot 0 = 0 \neq 1$.

3. This is equivalent to the statement that $a \cdot b = 0$ implies a or $b = 0$ iff for all $a \in S$ there exists a^{-1} such that $a \cdot a^{-1} = 1$.

First, assume that a^{-1} exists for all $a \in S$. Now, assume $ab = 0$, and assume $a, b \neq 0$. Since they are both nonzero, a^{-1} and b^{-1} exist. This gives us $aba^{-1}b^{-1} = 0a^{-1}b^{-1}$, so $1 = 0$, which is a contradiction. Therefore at least one of a, b must be 0.

Now, assume that if $ab = 0$ then a or $b = 0$. Then assume $a_1b = a_2b = c$. Then

$$a_1b + (-a_2b) = 0,$$
$$(a_1 + (-a_2))b = 0.$$

Because of our initial assumption, either $a_1 = a_2$ or $b = 0$. This implies that either $b = 0$, or if $ab = c$ for some fixed c then a is unique.

Now, say S has n elements and call them $a_1, a_2, \ldots, a_n$. Multiplying them by b gives us $a_1b, a_2b, \ldots, a_nb$. Since each a_i is unique, each a_ib is unique as well. Since there are n values of a_ib and there are n elements in S, exactly one of them is 1. The a_i such that $a_ib = 1$ is the inverse of b.

6. Assume $\mathbb{F}$ is a subfield of $\mathbb{Q}$. Then it must satisfy all field axioms with the same operations of $\mathbb{Q}$.

Since $\mathbb{F}$ is a field, it must contain the additive identity, 0, and the multiplicative identity, 1 in $\mathbb{Q}$. By repeatedly adding 1, we obtain all positive integers. Also, 1 has an additive inverse, -1. By repeatedly adding -1, we obtain all negative integers. Therefore, $\mathbb{F}$ contains $\mathbb{Z}$.

Also, $\mathbb{F}$ must contain the multiplicative inverses of all non-zero elements it contains, so if $n \in \mathbb{F}$ then $\frac{1}{n} \in \mathbb{F}$. So $\mathbb{F}$ contains all fractions $\frac{a}{b}$ where $a, b \in \mathbb{Z}$ and $b \neq 0$, which are all the elements of $\mathbb{Q}$. Since $\mathbb{F}$ contains all elements of $\mathbb{Q}$, it follows that $\mathbb{F} = \mathbb{Q}$, and there is no proper nontrivial subfield of $\mathbb{Q}$.

7. Assume, for contradiction, that there is a field $\mathbb{F}$ satisfying $\mathbb{R} \subsetneq \mathbb{F} \subsetneq \mathbb{C}$. So there is an element $z \in \mathbb{F}$ so that $z \notin \mathbb{R}$ but $z \in \mathbb{C}$. This means z can be written as $z = a + bi$ where $a, b \in \mathbb{R}$ and b is nonzero.

Isolating i gives us $i = \frac{z-a}{b}$. Since $a, b \in \mathbb{R}$, a has an additive inverse and b has a multiplicative inverse. So $\frac{z-a}{b}$ exists in $\mathbb{F}$, so $i \in \mathbb{F}$.

Since $\mathbb{F}$ contains all elements of $\mathbb{R}$ and it contains i, it contains all elements that can be obtained by adding ai or $-ai$ to any real number where $a \in \mathbb{R}$. This gives us all numbers of the form $c_1 + c_2i$, which is all of $\mathbb{C}$. Therefore, $\mathbb{F} = \mathbb{C}$, which is a contradiction. So there is no field $\mathbb{F}$ satisfying $\mathbb{R} \subsetneq \mathbb{F} \subsetneq \mathbb{C}$.

8. We need to prove that the axioms of a field apply to $\mathbb{Z}/p\mathbb{Z}$. All the axioms except (M2) hold because they are properties of integers.

From problem 1.3, if a finite set along with two operations satisfy everything except (M2) and it is true that $ab = 0$ implies that a or $b = 0$. So it suffices to show that if $ab \equiv 0 \pmod{p}$ then either a or $b \equiv 0 \pmod{p}$. Assume that $a, b \not\equiv 0 \pmod{p}$. Then $ab \not\equiv 0 \pmod{p}$, which is a contradiction. So $\mathbb{Z}/p\mathbb{Z}$ is a field.

11. Since G is a group, it satisfies $(gh)k = g(hk)$. So if $g(hk) = (gh)(gk)$ then $(gh)k = (gh)(gk)$. Multiplying by the inverse of gh gives us $k = gk$ for all $g, k \in G$. This means g is the identity element. Since this is true for all $g \in G$, $G = \{e\}$.

12. Assume axioms (A1R) and (A1L) to be true. Then $g \cdot e_R = g$. If we let $g = e_L$, we get $e_L \cdot e_R = e_R$. Similarly, letting $g = e_R$ in $e_L \cdot g = g$ gives us $e_L \cdot e_R = e_L$. Therefore, $e_R = e_L \cdot e_R = e_L$, so $e_R = e_L$. This means (A1R) and (A1L) imply (A1).

To go the other direction, assume there exists $e \in G$ so that for all $g \in G$, $g \cdot e = e \cdot g = g$. Then letting $e_R = e_L = e$ gives us $g \cdot e_L = e_R \cdot g = g$ which implies group axioms (A1R) and (A1L).